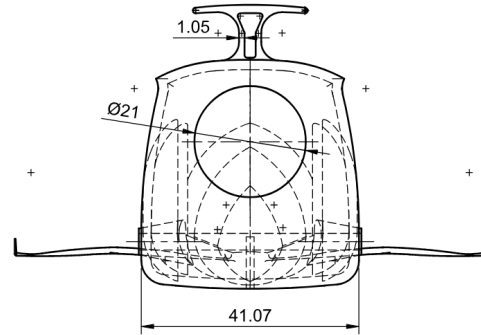
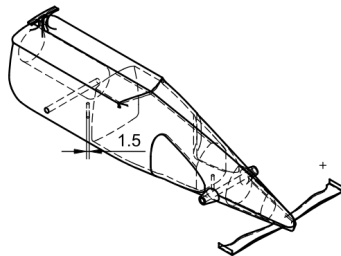
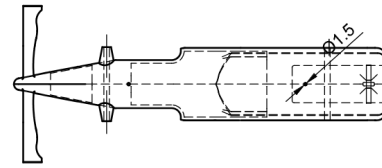
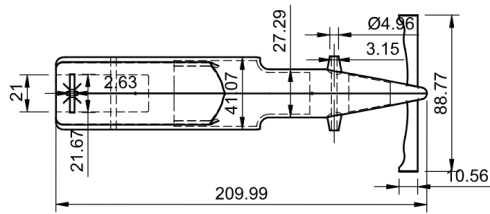




All measurements are in millimeters (mm).

Design Philosophy and Geometry

My dragster design features curvature and aerodynamic shaping that takes inspiration from the most aerodynamic shape, a raindrop. The entire design is made to be used as one single piece (excluding axles, wheels, and CO2 canister), and it can be easily 3D printed, carved out from wood, or CNC machined.

Aerodynamic Components

I have a front wing that helps to guide air, with its inspiration taken from the Red Bull RB8. This is the widest part of the entire design. The front wing was optimized to an 88.77 mm width to maximize aerodynamic surface area whilst maintaining a 1.23 mm safety buffer from the maximum width constraint of 90 mm.

I also implemented a rear wing, which allows for straight-line stability and helps air flow, thus reducing pressure drag. In a race that is won by thousands of a second, any sort of aerodynamic drag is crucial to be removed to the maximum extent allowed by the design and rules. The rear wing is also placed in a neutral position that does not allow for the downforce effect to take place, as that would also create aerodynamic drag (as seen in Formula 1 cars). I have two small rib-like extrusions going from the middle of the car to the rear. The function of these is to help guide air away from the rear of the car, where the CO2 canister is placed. Without something like this, there would be what is called “dirty air” going into the CO2 stream.

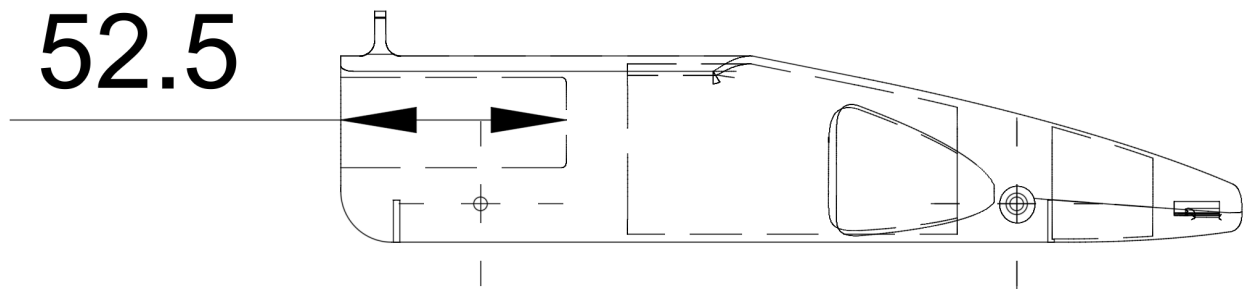
Dirty air is essentially air with a sort of vortex-like effect to it, which causes instability and unpredictability.

Compliance With TSA Regulations

To follow rules, I maintained a length of 210 mm (within the max 305 mm length rule), a width of ~89 mm (within the max 90 mm width rule), and I have a 5 mm minimum ground clearance to prevent the chassis from bottoming out and to also prevent the screw eyes from dragging on the track.

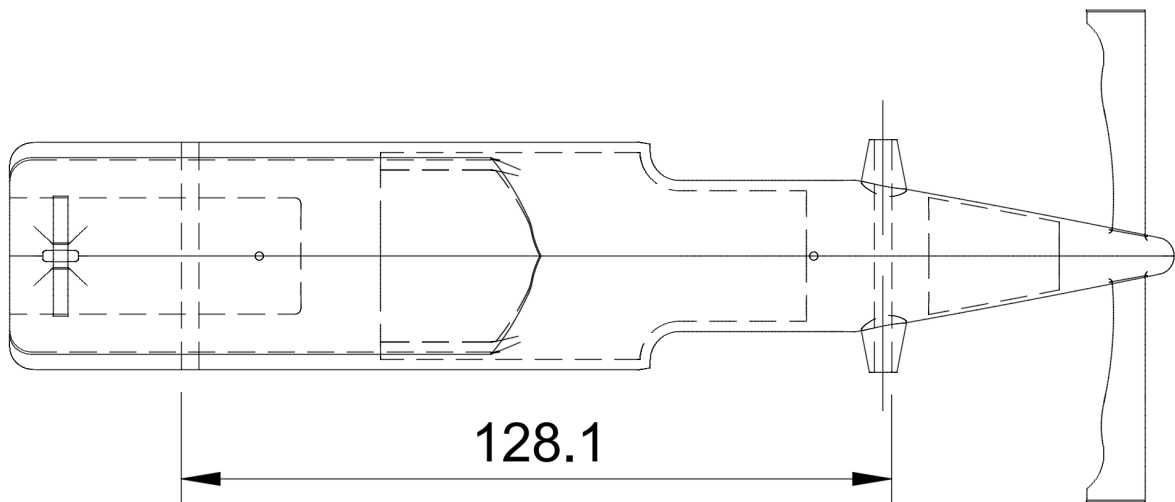
I ensured that for both the front and rear axles, the span is greater than 35 mm and also less than 42 mm. On top of this, none of the spaces for wheels are enclosed or wrapped around by the design, as it is not permitted. It is an entirely open-wheel design.

As for CO₂, I have maintained at least 5 mm of solid material around the entire CO₂ cartridge hole. My design also accommodates a hole depth within the 45 - 55 mm limit (~52.5 mm, pictured below).



The rib-like extrusions ensure that the exhaust path of the CO₂ canister is consistently clear, and I have made sure that the body does not interfere with the standard nozzle of the launch pod.

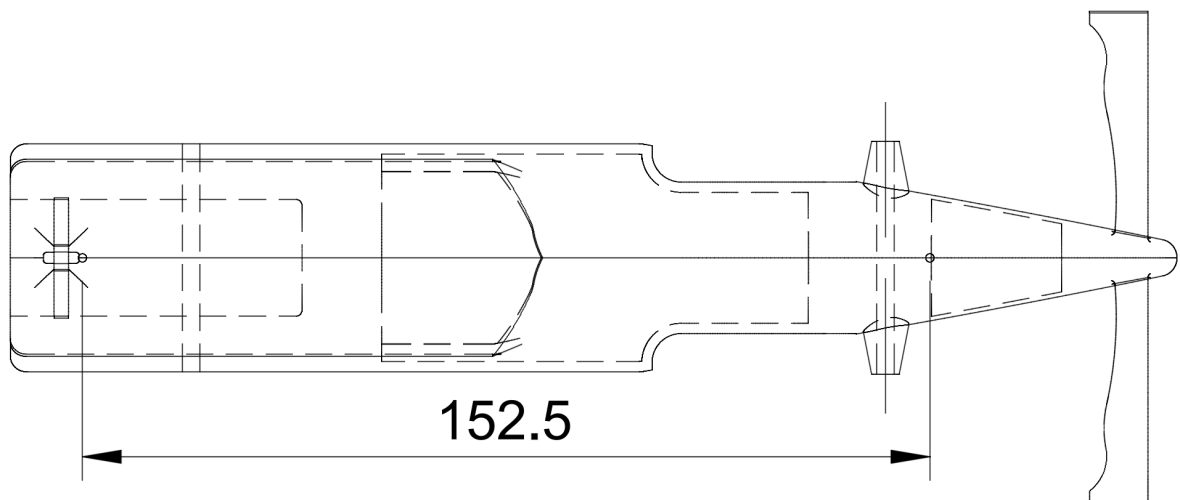
The distance between the center of my front axle and the center of my rear axle is between 105 mm and 270 mm (125.1 mm), as stated in the rules (pictured below).



Distance is listed as 125.1 mm and not 128.1 mm because axle hole (3 mm diameter) is factored in, as the distance between the center of both axles is being measured, not the distance between the farthest points.

I have targeted ~53 g of weight for the design, which is safely above the 40 g minimum. This ensures that the car is heavy enough to stay on the track, but light enough to ensure proper acceleration. I remained under the weight limit of 55 g, even when accounting for potential added weight from paint and hardware. I plan on 3D printing the design using LW PLA filament to ensure a lightweight car whilst also having structural integrity and proper density.

The holes for the screw eyes placed at the bottom of the design are 152.5 mm apart, which follows the rules (pictured below.)



Please note that minor changes were made to the design in comparison to the first images at the top of the document to aid in following the rule about the screw eye hole distance.

This provides a 2.5 mm safety margin to account for any potential shrinkage of material during the printing process, whilst also maintaining the car's stability.

Conclusion

Through integration of fluid dynamic principles and design knowledge, I created a design for a dragster car that fits the 2026 TSA regulations, while representing a highly optimized racing platform. I utilized a raindrop profile to lower pressure drag to a minimum and use guidance systems for stability, while also engineering the vehicle to maintain high velocity and minimum mechanical friction. My usage of LW PLA material creates a competitive, lightweight, and high strength chassis that remains within the regulatory threshold. This design is prepared for the manufacturing phase, and all critical components were verified and optimized for a successful and high-performance qualifying run.